

Deepjyoti Deka

Research Scientist

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Summary

Dr. Deka is a staff scientist and project leader working in Machine Learning for smart grids and cyber-physical networks. His research leverages network physics and statistics to design certifiable algorithms for network identification, grid control and decarbonization.

Date of Birth: 14 December 1987. **Citizenship:** India, **Permanent Resident:** USA

Education

2009 – 2015 **M.S./Ph.D. in Electrical & Computer Engineering**, *The University of Texas at Austin, TX*.

Dissertation: Analysis of the power grid: structure & secure operations.

Adviser: Prof. Sriram Vishwanath (Communication & Networks Track), CGPA: 3.94/4.

2005 – 2009 **B.Tech. in Electronics & Comm. Engineering**, *Indian Institute of Technology (IIT) Guwahati, India*. CGPA: 9.58/10, **Institute Silver medal** for highest graduating CGPA in department.

Awards and Achievements

2022 **Young Alumni Achiever Award**, IIT Guwahati, India.

2021 **Best Paper on Machine Learning Innovation** at ICML Workshop on Tackling Climate Change.

2015 **Best Conference Papers on Cyber Security, & Protection** at IEEE PES General Meeting.

2015 **Student Leadership Award** by School of Engineering, University of Texas Austin.

2009 **Institute Silver Medal** for best graduating student in ECE at IIT Guwahati.

2008 – 2009 **OP Jindal Engineering and Management scholarship** given to 25 students across India.

2006 – 2008 **Institute Merit Scholarship** for 06 – 07, 07 – 08 for first rank in the department at IIT Guwahati.

2003 Recipient of **NTSE merit scholarship** for higher education, awarded by the Government of India.

2003 Awarded **Gold Medal** for securing first position in regional mathematics olympiad.

Professional Experience

July 2024 – present **Research Scientist**, *MIT Energy Initiative, MIT*, Cambridge, Massachusetts,
Working on new data-driven approaches to resilient operation of grid resources and decarbonization.

July 2022 – **Scientist 3**, *Theoretical (T) Division, Los Alamos National Laboratory*, Los Alamos, New Mexico,

July 2024 **Mentoring:** 3 postdocs, 20+ summer students

Technical Projects: (a) Machine Learning augmented optimization for power systems,
(b) Robust neural-networks for network change detection and fault localization,
(c) Control of interdependent infrastructure networks and natural systems for reliability.

September 2018–June 2022 **Scientist 2**, *Theoretical (T) Division, Los Alamos National Laboratory*, Los Alamos, New Mexico,

Technical Projects: (a) Physics-informed machine learning for Topology and Parameter estimation,
(b) Aggregated Control of flexible loads in power grids, (c) Learning for Cyber-Physical Systems.

Dec. 2015 – Sept. 2018 **Postdoctoral Researcher**, *Center for Non-linear Studies (CNLS), Los Alamos National Laboratory*, Los Alamos, New Mexico, **Mentor:** Michael Chertkov, Theory (T) division.

May 2014 – **Summer Intern**, *Theory Division, Los Alamos National Laboratory*, Los Alamos, New Mexico,

August 2014 **Project:** Non-intrusive algorithm design to learn topology and consumption profiles in power grids.

May 2013 – **Summer Intern**, *Electric Reliability Council of Texas (ERCOT)*, Taylor, Texas,

August 2013 **Project:** Storage incorporation into ERCOT through the Fast Responding Regulation Service Pilot.

May 2008 – **Summer Intern**, *Corporate R & D, Qualcomm Inc.*, San Diego, CA,
July 2008 **Project:** Modeling and improvement in page download times over wireless networks.

Publications (34-Journal, 50-Conference, extended list at Google Scholar)

- [1] M.S. Veedu, **D Deka**, M. Salapaka. Information Theoretically Optimal Sample Complexity of Learning Dynamical Directed Acyclic Graphs, *International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2024.
- [2] W. Li, **D. Deka**, R. Wang, and M. Paternina. Perturbation-Robust Neural Networks for Stochastic Power Grids, *IEEE Transactions on Artificial Intelligence*, 2023.
- [3] **D Deka**, H. Doddi, S. Talukdar, and M. Salapaka. Learning Linear Dynamical Systems under Non-white Excitation, *International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2022.
- [4] W. Li and **D. Deka**. Physics-Informed Graph Neural Networks for Robust Fault Location in Power Grids, in *ICML Workshop on Tackling Climate Change*, 2021. (**Best paper: ML Innovation**).
- [5] S. Talukdar, **D. Deka**, H. Doddi, D. Materassi, M. Chertkov, and M.V. Salapaka. Physics Informed Topology Learning in Networks of Linear Dynamical Systems, *Automatica*, 2020.
- [6] S Gupta, S Misra, **D. Deka**, and V. Kekatos. DNN-based Policies for Stochastic AC Optimal Power Flow, *Journal of Electric Power Systems Research (EPSR)*, 2022.
- [7] I. Mezghani, S. Misra, and **D. Deka**. Stochastic AC Optimal Power Flow: A Data-Driven Approach, *Journal of Electric Power Systems Research (EPSR), special issue on Power Systems Computation Conference (PSCC)*, 2020.
- [8] **D. Deka**, and S. Misra. Learning for DC-OPF: Classifying active sets using neural nets, in *IEEE Powertech*, 2019.
- [9] A. Hassan, **D. Deka**, M. Chertkov, and Y. Dvorkin. Data-Driven Learning and Load Ensemble Control, *Journal of Electric Power Systems Research (EPSR)*, 2020.
- [10] **D. Deka**, S. Vishwanath, and R. Baldick. One breaker is enough: hidden topology attacks on power grids, in *IEEE PES General Meeting*, 2015. (**best Conference papers on Cyber Security, Stability, and Protection**).

Project Leadership

- 2022 –2025 co-PI, **LANL Laboratory Directed Research & Development Grant on** Machine Learning for Non-convex Optimization, *Funding: 1 million USD*.
- 2021 – 2024 PI, **LANL Laboratory Directed Research & Development Grant on** Resilient operation of interdependent engineered networks and natural system, *Funding: 1 million USD*.
- 2020 –2023 LANL co-PI, **US Department of Energy Advanced Grid Modeling Grant on** Machine Learning for Stochastic Optimization, *Funding: 1.3 million USD*.
- 2020 –2023 LANL co-PI, **US Department of Energy Advanced Grid Modeling Grant on** Real-Time Control of Large-Scale Power Grids, *Funding: 2.5 million USD*.
- 2018 – 2021 co-PI, **LANL Laboratory Directed Research & Development Grant on** Statistical Machine Learning for Cyber-Physical Systems, *Funding: 900k USD*.
- 2018 – 2019 co-PI, **US Department of Energy GMLC Grant on** Machine Learning for Distribution Systems, *Funding: 240k USD*.

Software Skills

MATLAB, PYTHON, JULIA, GUROBI, IPOPT, POWERWORLD, Deep Learning Spec. (Coursera)

Professional Service

- 2020- Senior Member, IEEE
- 2021-2022 Associate Editor, IEEE Transactions on Smart Grid
- TPC IEEE Smartgridcomm 2018,2019,2020, PMAPS 2020, IEEE GlobalSIP 2016-2019, AISTATS 2022